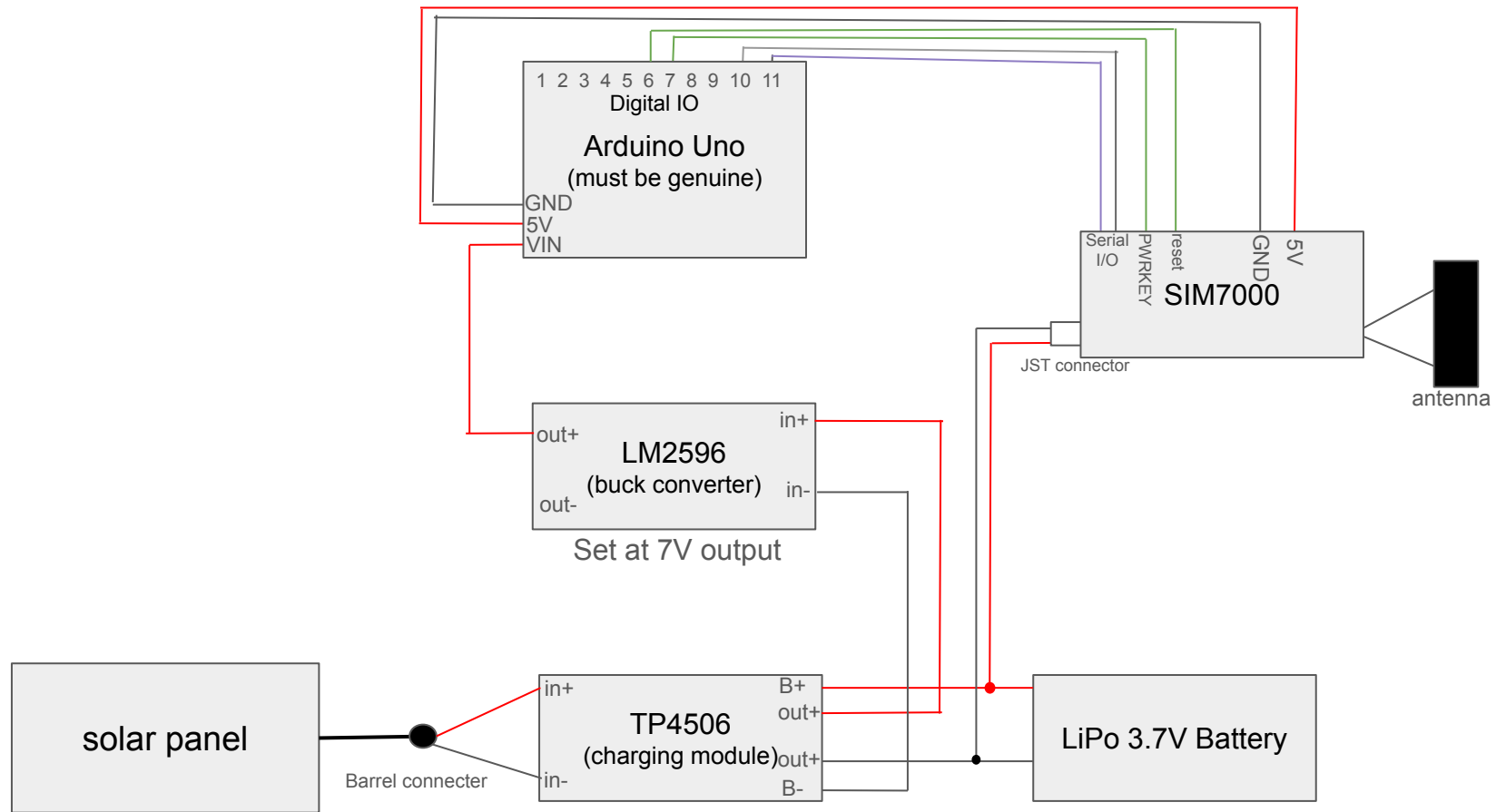


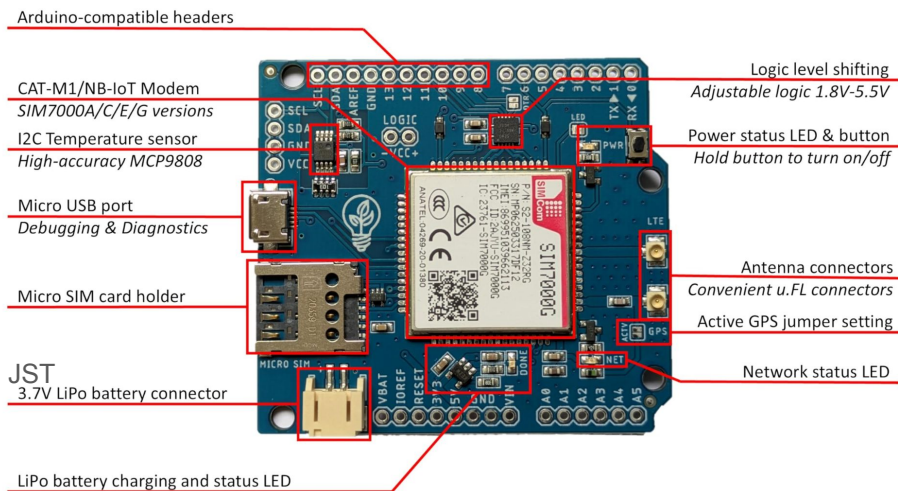


SIM7000 - Botletics - Hardware

This is an IC built on the SIMcom SIM7000 module, designed for easy use with an arduino. It can be soldered with header pins and mounted on an arduino uno or mega, but this was found to be problematic due to unneeded connections and issues with manufacturer quality. See [Official documentation](#). Note that this is incomplete. Botletics is run by one person who appears to be inactive. His [youtube](#) also has some content.

The SIM7000 only needs a few connections to function:

- JST connector wired directly to LiPo battery. It cannot be wired to the TP4506 (charging module) because the charging module will limit current to 1A. The SIM7000 may require up to a 2A pulse to transmit data. Most DC-DC converter (buck, boost, etc. like the LM2596) will also have too low of a current cap.
- Arduino 5V pin wired to SIM7000 5V pin. This is to ensure both components compute logic at the same voltage and can communicate over serial.
- GND on arduino to GND on sim7000. Same reasoning as the 5V pin.
- Pins 10/11 on the arduino wired to pins 10/11 on the SIM7000. These are used for serial (i.e., sending data and commands back and forth)
- Pin 6 on arduino to pin 6 on SIM7000. This is the power key, which can be pulled low for a few seconds to turn on the SIM7000. This allows the board to be turned on electrically, just as the physical PWR button (top right on the board) can be used to turn the board on manually.



- (Optional) Pin 7 on arduino to pin 7 on SIM7000. This is a reset pin which can be pulled low to reset the module. This is not strictly necessary but may be useful to help clear errors, etc.

Other notes:

- The SIM7000 also requires an antenna to send signals and a micro SIM card to be able to use data. A hologram SIM works best for this.
- Note that there are other methods to power the SIM7000. I found that none of these work besides the JST for providing sufficient power.

SIM7000 - Botletics - Software

Again, see official documentation. Go to examples and look, especially, at LTE demo. However, in this setup, the example code did not work.

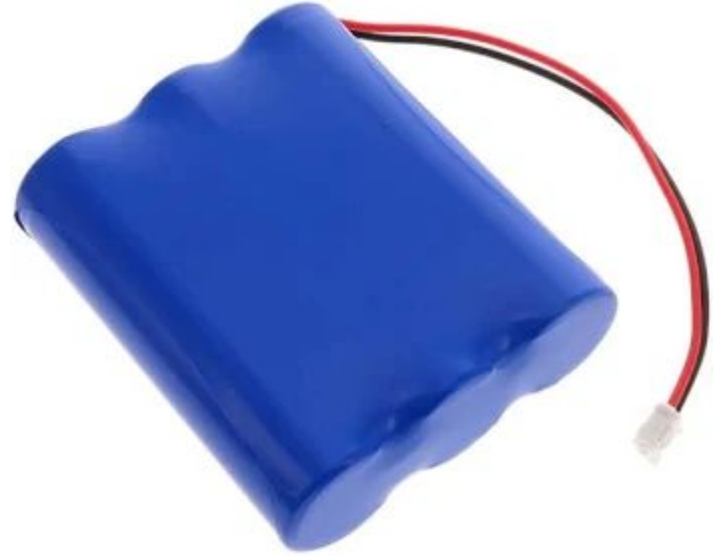
This application of the SIM7000 in this case, fundamentally, simply requires the SIM7000 to send an HTTP request to a server. There are a variety of ways to do this. The SIM7000 is a standard cellular module in that it uses antenna commands to function; i.e., commands beginning with “AT+” are sent over serial at a specified baud rate to interface with it. The SIM7000 defaults to 115200 baud, which is too high for the arduino to consistently use over a software serial. The command AT+IPR=9600 must be sent at 115200 baud initially to set the SIM7000’s baud rate at 9600, which the Arduino can comfortably work with.

A series of AT commands are then sent. These are standard for SIMCOM chips and can be referenced online. Note that the HTTP family of AT commands works best for opening and using an HTTP connection with the SIM7000. Below is a demo GET request using these commands.

```
sendCommand("AT+HTTPIPINIT", 1000);  
sendCommand("AT+HTTTPARA=\\"CID\\" ,1", 1000);  
sendCommand("AT+HTTTPARA=\\"URL\\" ,\\"http://httpbin.org/get\\" , 1000);  
sendCommand("AT+HTTTPACTION=0", 1000);  
sendCommand("AT+HTTTPREAD", 1000);  
sendCommand("AT+HTTPTERM", 1000);
```

3.7V LiPo Battery

- Any standard LiPo will work. As mentioned, needs to be able to deliver at least 2000mA instant at max. Must be rechargeable.
- For a standard 3.7V LiPo battery, if it left uncharged for long enough, it will degrade and stop working. Check with a multimeter - if the voltage is below 3V, the battery is degraded and non-functional.



Arduino

Basic understanding of the arduino is assumed. A few specific notes for this implementation:

- Genuine Arduino Unos must be used. After testing, the SIM7000 does not function with any 3rd party manufacturers. An Arduino Mega may be usable, but is overkill.
- The Arduino shares a ground with the SIM7000 and the LiPo. Do not connect the LM2596 OUT- with the arduino ground.

